

| Test Case                       | Input                             | Expected Output                                                                       | Description                                                                                        |
|---------------------------------|-----------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Insert node into BST            | Insert ("Govardhan", "555-12345") | None (no output, but node is added)                                                   | Inserting the first node into the tree.                                                            |
| Insert node into BST            | Insert ("Bob", "555-2345")        | None (no output, but node is added)                                                   | Inserting a second node. Bob should be placed as the left child of Govardhan.                      |
| Insert node into BST            | Insert ("Charlie", "555-3456")    | None (no output, but node is added)                                                   | Inserting Charlie; Charlie should be placed as the right child of Bob.                             |
| Insert node into BST            | Insert ("Alice", "555-1234")      | None (no output, but node is added)                                                   | Inserting Alice; Alice should be placed as the left child of Bob.                                  |
| Search for existing node        | Search for "Govardhan"            | "555-12345"                                                                           | Govardhan exists in the tree; it should return Govardhan's phone number.                           |
| Search for existing node        | Search for "Alice"                | "555-1234"                                                                            | Alice exists in the tree; it should return Alice's phone number.                                   |
| Search for non-existing node    | Search for "David"                | "Name not found"                                                                      | David does not exist in the tree, so it should return "Name not found."                            |
| In-order traversal              | Perform in-order traversal        | ["Alice - 555-1234", "Bob - 555-2345", "Charlie - 555-3456", "Govardhan - 555-12345"] | Performing in-order traversal should list all nodes in sorted order by name.                       |
| Delete an existing node         | Delete "Charlie"                  | None (no output, but Charlie is removed)                                              | Deletes the node Charlie from the tree.                                                            |
| In-order traversal after delete | Perform in-order traversal        | ["Alice - 555-1234", "Bob - 555-2345", "Govardhan - 555-12345"]                       | Performing in-order traversal after deleting Charlie. Charlie should no longer appear in the list. |